

IEOR E4101 Probability, Statistics and Simulation — Practice Midterm (Alt A)

Instructor: (same as original)

Instructions: Show your work. Justify any formulas you use.

Problem 1 (Discrete probability)

You throw a fair 8-sided die (faces 1–8) eight times.

- (a) What is the probability that all eight face values appear?
- (b) What is the probability that one or more 8's appear?

Problem 2 (Bayes' rule, conditional probability)

A 100%-reliable sonogram reveals a couple will have **twin girls**. In the population, the proportion of **identical** twins is **40%**. For identical twins, it is equally likely to be two boys or two girls. For fraternal twins, the sexes are independent and equally likely. What is the probability the couple's girls are **identical**?

Problem 3 (Scenario returns, correlation, portfolio risk)

There are three equally likely economic scenarios: *strong*, *normal*, *weak*. Next year's returns (A, B) under each scenario are:

Scenario	Stock A	Stock B
Strong	28%	15%
Normal	9%	−8%
Weak	−10%	8%

- (a) Compute $E[A]$ and $E[B]$.
- (b) Compute $\text{Corr}(A, B)$.
- (c) If you invest **50% in A and 50% in B**, compute the portfolio's **expected return** and **standard deviation**.

Problem 4 (Overbooking as a binomial)

An airline knows **7%** of passengers *do not* show up. The plane holds **80** passengers, and the airline sells **82** tickets. Let X be the number who show. What is $P(\text{everyone gets a seat}) = P(X \leq 80)$?

Problem 5 (Records in i.i.d. continuous sequences)

Let $X_1, X_2, \dots$ be i.i.d. continuous. Say a **record low** occurs at time n if $X_n < \min(X_1, \dots, X_{n-1})$. Let Y_n be the number of record lows by time n (and assume a record at time 1).

- (a) Find $E[Y_n]$ for general n .
- (b) Find $\text{Var}(Y_n)$ for $n = 2$ and $n = 3$.

Problem 6 (Expected profit with a triangular density)

An item's weight X (ounces) has density

$$f(x) = \begin{cases} x - 7, & 7 \leq x \leq 8, \\ 9 - x, & 8 \leq x \leq 9, \\ 0, & \text{otherwise.} \end{cases}$$

The selling price is \$2.40. A full refund is given if $X < 7.50$ oz. Production cost is $c(x) = x/12 + 0.30$. Compute the **expected profit per item**.